

UNIwersytet VIZJA • SCHOOL OF COMPUTER SCIENCE & TECHNOLOGIES

Design and Implementation of a Web-Based Student Management System for Educational Institutions



VIZJA UNIVERSITY

LIVE <https://thesms.me>

STUDENT

Maksym Shpak

Index: 45567

SUPERVISOR

Marcin Kacprowicz

Diploma seminar

FIELD OF STUDY

Computer Science

Engineering degree • 2026

Presentation Agenda

- | | |
|--|--|
| <p>01 Problem Statement & Purpose
Spreadsheets, paper registers, no audit trail</p> <hr/> | <p>05 Database Design
6 models, 6 migrations, constraints in the schema</p> <hr/> |
| <p>02 Comparative Analysis
Moodle, USOS, Google Classroom — and the gap</p> <hr/> | <p>06 Key Features
Weighted grades, RBAC, EN/PL, 96 tests</p> <hr/> |
| <p>03 Technology Stack
Django 6, Python 3.12, PostgreSQL 18</p> <hr/> | <p>07 Application Screenshots
Dashboard, student records, mobile layout</p> <hr/> |
| <p>04 System Architecture
Browser › Render TLS › Gunicorn › PostgreSQL</p> | <p>08 Deployment & Results
Live on Render + Neon, then what comes next</p> |

Problem Statement & Purpose

3 places

spreadsheet, paper register
and email thread

0

audit trail of who entered
or edited a grade

by hand

every average and every
attendance total

THE PROBLEM

- **Fragmented records** — spreadsheets, paper registers, email threads
- **No single view** — grades and attendance never meet in one place
- **Manual arithmetic** — averages recalculated by hand, errors slip through
- **No accountability** — nobody can prove who recorded a mark
- **No confidentiality** — every teacher can see every student
- **No Polish interface** — staff work in English-only tools

THE PURPOSE

- **One source of truth** — a relational schema with enforced constraints
- **Instant analytics** — KPI tiles and three charts from ORM aggregation
- **Automatic authorship** — every grade stores who assigned it
- **Scoped visibility** — teachers see their own courses, admins see all
- **Bilingual by design** — the whole interface switches EN $\rightleftharpoons$ PL
- **Deployable** — one build script, HTTPS, managed PostgreSQL

Result: one auditable, role-aware platform now managing 25 students, 8 courses, 87 enrollment records — 51 of them active — and 87 grades in production.

Comparative Analysis

CRITERION	MOODLE	USOS	GOOGLE CLASSROOM	SMS — THIS PROJECT
Target user	Large universities	Polish public HEIs	Schools and courses	Departments and small institutions
Deployment	Self-hosted, heavy	Vendor-managed	Google cloud only	Self-hosted, one Django app
Setup effort	Days of plugin work	Institutional contract	Minutes, no control	Minutes — a single build script
Gradebook	Yes, complex	Yes	Basic	Weighted components + author audit
Attendance register	Plugin required	Yes	Not available	Built in, three statuses
Role isolation	Configurable, intricate	Yes	Limited	Enforced at queryset level
Bilingual EN / PL	Language packs	Polish-first	Yes	Native, 312 strings
Analytics	Reports plugin	Institutional reports	Minimal	Chart.js dashboard + CSV
Cost of ownership	GPL + hosting	Licensed	Free, data in Google	Open source, €0 hosting

Positioning: the established platforms are either too heavy or too closed for a single department; SMS trades breadth for a focused, auditable and fully self-hosted core.

Technology Stack

BACKEND

Runtime · data · authentication

- Python 3.12
- Django 6.0.6
- PostgreSQL 18 (Neon)
- SQLite 3 for development
- Django session authentication
- Custom user model with roles
- Django ORM · 6 migrations
- dj-database-url 2.1

FRONTEND

Templates · charts · theming

- Django Templates · 30 files
- Semantic HTML5
- CSS custom properties · 1 945 lines
- Vanilla JavaScript ES6 · 425 lines
- Chart.js 4 — doughnut and bars
- Turbo 8 page transitions
- Bootstrap Icons
- Light and dark themes

INFRASTRUCTURE

Hosting · operations · CI

- Render web service
- Neon managed PostgreSQL
- Gunicorn 22 (WSGI)
- WhiteNoise 6.6 static files
- HTTPS with HSTS preload
- GitHub Actions CI
- Git version control
- Zero CDN dependencies

No build step, no separate API server. Every asset is vendored locally, so the application runs and demonstrates fully offline.

System Architecture

CLIENT

Browser

HTML · CSS · ES6

Turbo navigation

EDGE

Render · TLS 443

HTTPS, HSTS

static via WhiteNoise

APPLICATION

Gunicorn + Django 6

URL › View › Template

ORM aggregation

DATA

PostgreSQL 18

Neon managed

DATABASE_URL, one codebase

REQUEST PIPELINE — MIDDLEWARE

Security

Session

CSRF

Locale

WhiteNoise

Authentication

Role decorators

Queryset scoping

SUPPORTING SERVICES

i18n EN / PL

Interface language switch

GitHub Actions

Tests and deploy gates

Teachers screen

Admin issues lecturer accounts

CSV export

Role-scoped reporting

/api/dashboard/

JSON metrics endpoint

Database Design

User *extends AbstractUser*

id
username • email
first_name • last_name
role → admin | teacher
password (PBKDF2)
is_active

Student *academic identity*

id
student_number (unique)
first_name • last_name
email
study_program
date_enrolled • is_active

Course *taught unit*

id
course_code (unique)
course_name • description
semester • credits
max_students
teacher_id → User

Enrollment *student ↔ course*

id
student_id → Student
course_id → Course
enrollment_date
status → active | completed | dropped
unique (student, course)

Grade *weighted component*

id
enrollment_id → Enrollment
kind → coursework | midterm | final | retake
weight 1-100 %
grade_value 2.0 – 5.0
assigned_by → User

Attendance *daily register*

id
enrollment_id → Enrollment
date
status → present | absent | late
notes
unique (enrollment, date)

Integrity where it belongs: unique constraints, foreign keys and cascade rules live in the schema, so no interface path can create a duplicate enrollment or an orphaned grade.

Key Features

Academic Records

CRUD for students, courses and enrollments.
Administrators issue lecturer accounts on Teachers.
Enrollment is refused once max_students is reached.

Weighted Gradebook

Coursework, midterm, exam and retake — each with a weight. The course mark is their weighted mean. assigned_by records who last saved the component.

Attendance Register

Present, absent and late — one row per enrollment per day, enforced in the schema. Mark Class saves a whole roster in a single request.

Analytics & Export

Four KPI tiles, three Chart.js views and JSON at /api/dashboard/. CSV export of students and grades is scoped to the caller's role.

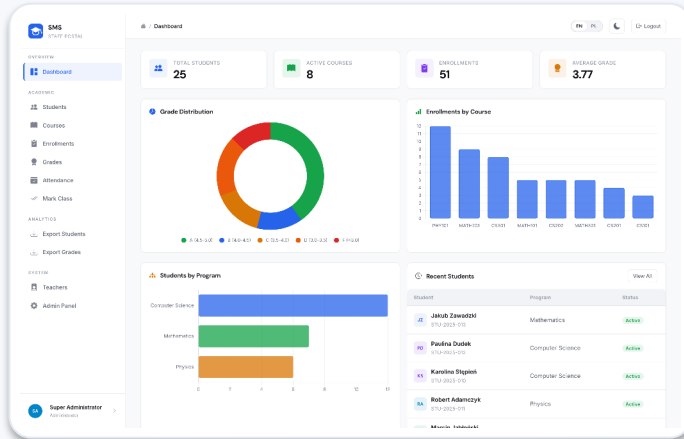
Role-Based Access

Administrators see the institution; teachers see only their courses. Decorators guard the route, querysets filter again — a guessed URL returns nothing.

i18n & Quality

312 strings, English $\rightleftharpoons$ Polish, locale-aware dates. 96 automated tests and four CI gates (tests, migrations, deploy check, collectstatic) on every push.

Application Screenshots



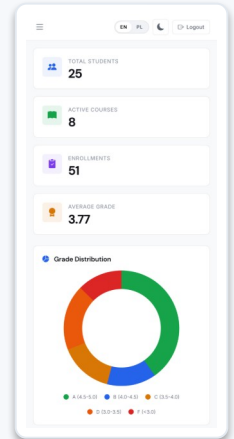
Dashboard — KPI tiles and three Chart.js views

The student register displays a table of students with columns for Student, Student Number, Program, Enrolled, Courses, Avg. Grade, Status, and Actions. The table is filtered by 'All Programs' and 'All Status'. A search bar is available at the top. The bottom right shows a list of recent students.

Student register — search, filters, pagination

The Mark Class page displays a table of students with columns for Student, Student Number, and Attendance. The table is filtered by 'All Records'. A search bar is available at the top. The bottom right shows a list of recent students.

Mark Class — a whole roster saved in one request



Responsive
down to 375 px

Every table is role-aware: a teacher opening the same page sees only students in their own courses, and the action buttons disappear.

Deployment & Results

 **LIVE** <https://thesms.me>

DEFENCE LOGIN (teacher)
prof.martinez / demo1234

25

students

8

courses

87

enrollment records

87

grades

€0

monthly cost

DELIVERED

- **Live on Render + Neon** — HTTPS, HSTS, managed PostgreSQL
- **Six-model schema** — constraints and an audit trail in the database
- **Role isolation** — admins and teachers see strictly their own data
- **Bilingual interface** — 312 strings, locale-aware dates and numbers
- **96 tests, four CI gates** — green on every push, then build.sh deploys

FUTURE WORK

- **Student portal** — a third role with read-only access to own results
- **Semester transcripts** — generated PDF reports per student
- **Email notifications** — alerts on new grades and absences
- **Documented REST API** — token-authenticated public interface
- **Two-factor login** — TOTP for administrator accounts

Thank you for your attention

Questions are welcome.

Maksym Shpak

Computer Science • Index: 45567

<https://thesms.me>



VIZJA UNIVERSITY